Chapter 1: The Real and Complex Number Systems

Section: Euclidean Spaces

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Definitions and Setup

Theorem 1.37: Properties of the Norm

Proof of Theorem 1.37

Remarks

Summary

Definitions and Setup

Why Euclidean Spaces?

- So far: $\mathbb{R}$ (the real line) and $\mathbb{C}$ (the complex plane)

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- Many applications need higher dimensions: physics, geometry, data
- Goal: define $\mathbb{R}^k$ with an inner product and norm, then prove key inequalities

Definition 1.36: The Space $\mathbb{R}^k$

Definition 1.36: Points and Vectors

For each positive integer k, let $\mathbb{R}^k$ be the set of all ordered k-tuples

$$\mathbf{x} = (x_1, x_2, \ldots, x_k),$$

where $x_1, \ldots, x_k$ are real numbers, called the coordinates of $\mathbf{x}$.

The elements of $\mathbb{R}^k$ are called *points* or *vectors*.

Definition 1.36: Vector Operations

Let $\mathbf{x} = (x_1, \ldots, x_k)$, $\mathbf{y} = (y_1, \ldots, y_k)$, and $\alpha \in \mathbb{R}$.

Addition:

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_k + y_k)$$

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Scalar multiplication:

$$\alpha\mathbf{x} = (\alpha x_1, \ldots, \alpha x_k)$$

Definition 1.36: Vector Space Structure

With these operations, $\mathbb{R}^k$ is a vector space over the real field.

The *zero element* (also called the *origin* or *null vector*):

$$\mathbf{0} = (0, 0, \ldots, 0)$$

Key: All vector space laws (commutativity, associativity, distributivity) follow immediately from the corresponding laws for $\mathbb{R}$.

Definition 1.36: Inner Product and Norm

Inner product (or scalar product) of $\mathbf{x}$ and $\mathbf{y}$:

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Inner product (or scalar product) of **x** and **y**:

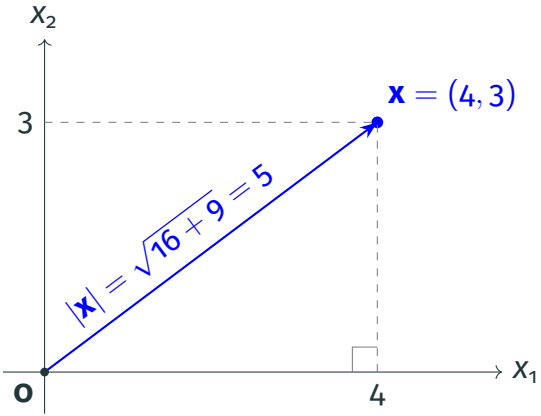
$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{k} x_i y_i$$

Norm of **x**:

$$|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^{k} x_i^2 \right)^{1/2}$$

The space $\mathbb{R}^k$ with this inner product and norm is called euclidean k-space.

Visualizing $\mathbb{R}^2$: Inner Product and Norm



Theorem 1.37: Properties of the Norm

Theorem 1.37: Norm Properties

Theorem 1.37

Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k$, and α is real. Then:

(a) $|\mathbf{x}| \geq 0$

(b) $|\mathbf{x}| = 0$ if and only if $\mathbf{x} = \mathbf{0}$

(c) $|\alpha \mathbf{x}| = |\alpha| \, |\mathbf{x}|$

Theorem 1.37: Norm Properties (continued)

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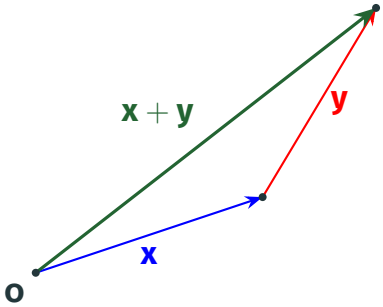
(f) $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$

The Triangle Inequality in $\mathbb{R}^2$



$$|\mathbf{x}+\mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$$

Proof of Theorem 1.37

Proof Strategy: Theorem 1.37

Goal: Prove all six properties of the norm.

Approach:

1. Parts (a), (b), (c): follow directly from the definition
2. Part (d): immediate from the Schwarz inequality (Theorem 1.35)
3. Part (e): expand $|\mathbf{x} + \mathbf{y}|^2$ and apply (d)
4. Part (f): follows from (e) by substitution

Proof of Parts (a), (b), and (c)

Parts (a), (b), (c)

Recall $|\mathbf{x}| = \left(\sum_{i=1}^{k} x_i^2\right)^{1/2}$.

(a) Each $x_i^2 \geq 0$, so $|\mathbf{x}| \geq 0$.

(b) $|\mathbf{x}| = 0 \;\Leftrightarrow\; \sum x_i^2 = 0 \;\Leftrightarrow\;$ each $x_i = 0 \;\Leftrightarrow\; \mathbf{x} = \mathbf{0}$.

(c) $|\alpha\mathbf{x}| = \left(\sum(\alpha x_i)^2\right)^{1/2} = \left(\alpha^2 \sum x_i^2\right)^{1/2} = |\alpha|\,|\mathbf{x}|$.

Proof of Part (d): Schwarz Inequality

Part (d): $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}|\,|\mathbf{y}|$

This is an immediate consequence of the Schwarz inequality (Theorem 1.35), which was proved in the previous section for sums of the form

$$\left|\sum a_j \bar{b}_j\right|^2 \leq \sum |a_j|^2 \cdot \sum |b_j|^2.$$

For real numbers, $\bar{b}_j = b_j$ and $|a_j|^2 = a_j^2$, giving exactly:

$$|\mathbf{x} \cdot \mathbf{y}|^2 \leq |\mathbf{x}|^2 \cdot |\mathbf{y}|^2.$$

Proof of Part (e): Triangle Inequality — Step 1

Step 1: Expand $|\mathbf{x}+\mathbf{y}|^2$

$$
\begin{aligned}
|\mathbf{x}+\mathbf{y}|^2 &= (\mathbf{x}+\mathbf{y})\cdot(\mathbf{x}+\mathbf{y}) \\
&= \mathbf{x}\cdot\mathbf{x} + 2\,\mathbf{x}\cdot\mathbf{y} + \mathbf{y}\cdot\mathbf{y} \\
&= |\mathbf{x}|^2 + 2\,\mathbf{x}\cdot\mathbf{y} + |\mathbf{y}|^2
\end{aligned}
$$

Proof of Part (e): Triangle Inequality — Step 2

Step 2: Apply the Schwarz inequality

By part (d): $\mathbf{x} \cdot \mathbf{y} \leq |\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}|\,|\mathbf{y}|$.

Therefore:

$$\begin{aligned} |\mathbf{x}+\mathbf{y}|^2 &\leq |\mathbf{x}|^2 + 2\,|\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^2 \\ &= \big(|\mathbf{x}| + |\mathbf{y}|\big)^2 \end{aligned}$$

Taking square roots: $\boxed{|\mathbf{x}+\mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|}$

Proof of Part (f): Distance Form

Part (f): Substitution trick

In part (e), replace $\mathbf{x}$ with $\mathbf{x} - \mathbf{y}$ and $\mathbf{y}$ with $\mathbf{y} - \mathbf{z}$:

$$|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$$

The left side simplifies: $(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z}) = \mathbf{x} - \mathbf{z}$.

Therefore: $\boxed{|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|}$ $\square$

Remarks

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Theorem 1.37 parts (a), (b), and (f) show that $\mathbb{R}^k$ can be regarded as a metric space, with distance $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$.

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Special cases:

- $\mathbb{R}^1$ = the *real line*; the norm is the absolute value $|x|$
- $\mathbb{R}^2$ = the *plane* (= the *complex plane*); the norm is the modulus $|z|$

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- $\mathbb{R}^k$ is the set of ordered k-tuples; a vector space over $\mathbb{R}$
- The inner product $\mathbf{x} \cdot \mathbf{y} = \sum x_i y_i$ and norm $|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2}$ give $\mathbb{R}^k$ its geometry
- The Schwarz inequality and triangle inequality (Theorem 1.37) are the key results
- $\mathbb{R}^k$ is a metric space, unifying $\mathbb{R}^1$, $\mathbb{R}^2 \cong \mathbb{C}$, and higher dimensions

Coming up next: The Appendix — constructing $\mathbb{R}$ from $\mathbb{Q}$ via Dedekind cuts.